Highlights

The Transformation of Broadband Demand: From Discretionary Service to Essential Infrastructure (2010–2024)

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- Broadband price elasticity declined from -0.60 (EaP) and -0.10 (EU) to ≈ 0 by 2020
- EaP price elasticity $5.9\times$ EU pre-COVID; demand-side harmonisation achieved by 2020
- Transformation began in 2015—not 2020—ruling out COVID-19 as the sole causal shock
- GNI-based prices give significant results in all specs; PPP in only 12% of EaP specs
- Policy must shift from price subsidies to infrastructure deployment as demand matures

The Transformation of Broadband Demand: From Discretionary Service to Essential Infrastructure (2010–2024)

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Abstract

Has broadband become a necessity good immune to price changes? Using a 15-year panel of 33 European countries (2010–2024) and two-way fixed effects with Driscoll–Kraay standard errors, we document a fundamental transformation in broadband demand. Pre-COVID, Eastern Partnership countries exhibited highly elastic demand ($\varepsilon = -0.60$, $p<0.001$)—a 10% price reduction increased subscriptions by 6.0%—while EU countries showed moderate elasticity ($\varepsilon = -0.10$, $p<0.10$). By 2020–2024, both regions converged to near-zero elasticity, with price changes having no detectable effect on adoption. Crucially, placebo tests reveal this transformation began in 2015, not 2020, indicating a decade-long digital integration process rather than a COVID-19 shock. We further demonstrate that price measurement critically affects inference for EaP countries: income-relative prices (as % of GNI) yield

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significant EaP elasticities in all specifications, whereas PPP-adjusted prices yield significant results in only 12% of EaP specifications; for EU countries, all three price measures perform broadly similarly. These findings have immediate policy relevance: as broadband transitions from discretionary service to essential utility, policy emphasis must shift from affordability subsidies to universal infrastructure deployment.
Keywords: Broadband demand, Price elasticity, Panel data, Eastern Partnership, European Union, COVID-19, Digital transformation

1. Introduction

When does a technology transition from luxury to necessity? For broadband internet, this question has profound implications: if demand becomes price-inelastic, subsidies and price regulations—the dominant policy tools of the past two decades—lose their effectiveness. The transformation of broadband from discretionary communication service to essential infrastructure for economic participation, education, healthcare, and civic engagement (Katz et al., 2010; Bertschek and Niebel, 2016) suggests this transition may already be underway. Yet empirical evidence on whether and when broadband demand became price-insensitive remains surprisingly scarce, despite substantial public investments in broadband infrastructure worldwide (OECD, 2020).

This paper provides such evidence by examining how broadband price elasticity evolved over 2010–2024 in 33 countries: 27 European Union (EU) member states and 6 Eastern Partnership (EaP) countries. We leverage substantial cross-country and temporal variation in broadband prices and adop-

tion rates to estimate time-varying elasticities using two-way fixed effects models. Our analysis reveals a striking structural transformation: Eastern Partnership countries exhibited highly price-elastic demand ($\varepsilon = -0.60$) during 2010–2019, but transitioned to price-inelastic demand ($\varepsilon \approx 0$) by 2020–2024, indicating broadband's evolution from discretionary service to essential necessity.

1.1. Research Questions and Motivation

Three core questions motivate this research. First, *how has broadband demand elasticity evolved over the past decade and a half?* Existing literature predominantly estimates elasticities at single points in time (Grzybowski, 2015b; Madden and Coble-Neal, 2015), yet technological change and digital economy expansion suggest elasticity may be time-varying as services transition from luxury to necessity (Hausman et al., 2001).

Second, *what explains regional heterogeneity in price sensitivity?* Eastern Partnership countries—comprising Armenia, Azerbaijan, Belarus, Georgia, Moldova, and Ukraine—represent lower-income markets where affordability concerns might generate higher price elasticity compared to wealthier EU markets. Understanding this heterogeneity is crucial for tailoring policy interventions to different development contexts (Waverman et al., 2001).

Third, *did COVID-19 fundamentally alter broadband demand?* The pandemic forced abrupt transitions to remote work, online education, and digital service delivery (OECD, 2021b), potentially transforming broadband from convenience to necessity overnight. However, distinguishing COVID's causal impact from pre-existing trends requires careful econometric identification, which we address through placebo tests.

1.2. Contributions

This paper makes four principal contributions to telecommunications economics and policy.

First, we document time-varying demand elasticity across a 15-year panel, revealing broadband's structural transformation. Unlike prior single-period estimates, our year-by-year analysis shows elasticity declined gradually from 2015 onward, not suddenly in 2020. This challenges the narrative of COVID-19 as an exogenous shock and instead suggests a decade-long evolution driven by digital economy expansion.

Second, we demonstrate that price measurement critically affects empirical inference. Using three alternative price measures—price as percentage of GNI per capita (income-relative), PPP-adjusted prices, and nominal USD prices—we show only income-relative prices yield consistent, statistically significant elasticity estimates. PPP-adjusted prices, commonly used in cross-country comparisons, produce estimates significant in just 12% of EaP specifications (at $p<0.05$). This methodological contribution has implications beyond broadband demand, highlighting the importance of price measurement in technology goods research where standard PPP baskets may not apply (Schreyer, 2002).

Third, we provide robust evidence of regional heterogeneity in price sensitivity. EaP elasticity was 5.9 times larger than EU elasticity during 2010–2019, consistent with income effects being stronger in developing markets (Röller and Waverman, 2001). This regional gap diminished during COVID-19, with both regions converging to near-zero elasticity—constituting empirical evidence that demand-side digital harmonisation between EaP and EU

markets is not merely a policy aspiration but an observable outcome.

Fourth, we offer policy-relevant insights on the effectiveness of affordability interventions versus infrastructure investment. As broadband transitioned to necessity status, price-based interventions lost effectiveness while infrastructure investment and quality standards became more important. This has implications for digital inclusion policies worldwide (Guermazi et al., 2021).

1.3. Preview of Findings

Our analysis yields three main findings. Pre-COVID elasticities differ sharply by region, with EaP countries 5.9 times more price-sensitive than EU countries—a result robust across 8 control specifications. Both regions then converge to near-zero elasticity during 2020–2024. Crucially, placebo tests show this decline began around 2015, pointing to secular digital integration rather than a COVID-19 shock.

The remainder of this paper proceeds as follows. Section 2 reviews relevant literature on telecommunications demand, digital transformation, and panel data methods. Section 3 describes our data sources and sample construction. Section 4 presents our empirical strategy. Section 5 reports baseline and robustness results. Section 6 discusses policy implications and limitations. Section 7 concludes.

2. Literature Review and Theoretical Framework

2.1. Broadband Demand and Price Elasticity

The telecommunications demand literature has extensively studied price elasticity, though predominantly for voice telephony and mobile services

(Hausman et al., 2001; Madden and Coble-Neal, 2015). Early broadband studies estimated elasticities ranging from -0.2 to -0.8, with substantial variation across methodologies and contexts (Rappoport et al., 2003; Hauge and Prieger, 2010). Grzybowski (2015b) estimate elasticities of -0.29 to -0.45 for European countries during 2007–2012, finding higher elasticity in lower-income countries. Our EaP estimate is consistent with this pattern, extending evidence to 2010–2024.

A key theoretical insight is that demand elasticity varies with necessity status. As connectivity integrates into economic and social life, price sensitivity declines as services shift from discretionary to essential (Hausman et al., 2001). Brynjolfsson et al. (2003) show similar patterns for information goods, where high initial elasticity declines as network effects and complementarities increase adoption. Our finding of declining elasticity from 2015–2024 aligns with this theoretical framework.

Recent work emphasizes time-varying elasticity in technology markets. Evidence from broadband markets suggests that price sensitivity declines as services become embedded in daily economic life and as complementary technologies (smartphones, streaming, cloud applications) raise the opportunity cost of disconnection (Goolsbee and Klenow, 2006). Our contribution extends this literature by documenting elasticity evolution across a 15-year panel with explicit pre-trend testing.

Several studies published in this journal have examined the *intensive margin* of broadband demand—consumers' willingness to pay for higher speeds or better quality conditional on being connected (Liu et al., 2018; Lindlacher, 2021; Sinclair, 2023; Grzybowski et al., 2014). Our paper complements this

strand by focusing on the *extensive margin*: the adoption decision itself. Whether a household subscribes at all is governed by price relative to income and infrastructure availability—both of which we measure directly. The regional comparison of EU versus EaP countries provides heterogeneity that single-country intensive-margin studies cannot capture: income constraints bind far more tightly in EaP markets, so extensive-margin elasticity is substantially higher, while both regions converge to near-zero elasticity as broadband transitions from discretionary to essential service.

2.2. Regional Heterogeneity and Development Context

Cross-country telecommunications studies consistently find higher price elasticity in developing markets (Waverman et al., 2001; Katz et al., 2010). Röller and Waverman (2001) show income effects dominate technology adoption in lower-income countries, where affordability constraints bind more tightly. Bertschek and Niebel (2016) find similar patterns for broadband, with elasticity inversely related to GDP per capita.

Our analysis focuses on Eastern Partnership countries—Armenia, Azerbaijan, Belarus, Georgia, Moldova, and Ukraine—which represent middle-income markets with substantial development heterogeneity relative to the EU (Ecorys, 2013). Critically, this grouping is not a researcher's sampling choice but an institutionally and politically defined set, established under the EU's Eastern Partnership policy framework and subsequently operationalised through the EU4Digital initiative, which targets precisely these six countries for telecommunications regulatory harmonisation and progressive integration into the European Digital Single Market (European Commission, 2021). The research question examined in this paper—whether demand-side conver-

gence to EU necessity-good status is feasible over a policy-relevant horizon—is meaningful only for this institutionally defined group, as it concerns a specific integration trajectory rather than a random draw from a broader population. Consequently, N=6 reflects the exhaustive membership of the Eastern Partnership, not a statistical sample: there is no broader population of EaP countries from which additional observations could be drawn, and adding non-EaP transition economies would conflate distinct policy regimes. Limited prior work examines broadband demand elasticity in this region (Koutroumpis, 2009), making the present analysis a direct contribution to an understudied policy context.

2.3. COVID-19 and Digital Transformation

The pandemic's impact on telecommunications demand has received substantial attention, though primarily descriptive (OECD, 2021b; IAB, 2020). Bökermann and Scheuermann (2021) document 30–50% traffic increases during March–April 2020, while Favale et al. (2020) show usage patterns shifted dramatically toward videoconferencing and streaming. However, causal evidence on price elasticity changes remains limited.

Theoretically, COVID-19 could reduce elasticity through two mechanisms. First, lockdowns and social distancing converted broadband from convenience to necessity for remote work and education (Brynjolfsson et al., 2020). Second, once connectivity became non-negotiable for school and employment, households continued subscribing even as incomes fell—treating broadband the way they treat electricity, so that the usual price-cutting response to financial stress no longer applied (Dietrich et al., 2020). Our contribution is econometric identification of elasticity changes using panel methods rather

than descriptive evidence.

Importantly, several studies suggest digital transformation predated COVID-19. Goldfarb and Tucker (2019) argue the pandemic accelerated existing trends rather than creating new ones, consistent with survey evidence on the rapid shift to remote work (Brynjolfsson et al., 2020). This aligns with our placebo test evidence showing elasticity trends began before 2020, suggesting gradual evolution rather than pandemic-induced shock.

2.4. Econometric Methods for Panel Data

Our empirical strategy builds on panel data econometrics with multiple dimensions of heterogeneity (Wooldridge, 2010; Baltagi, 2021). Two-way fixed effects models—with country and time effects—are standard for cross-country telecommunications studies (Koutroumpis, 2009; Czernich et al., 2011), as they control for time-invariant country characteristics and common time shocks while identifying from within-country variation.

A critical methodological issue is inference with panel data exhibiting cross-sectional dependence and serial correlation. Standard clustered standard errors may be inadequate when common shocks (e.g., financial crises, COVID-19) affect all countries (Cameron and Miller, 2015). Driscoll and Kraay (1998) propose kernel-based standard errors robust to heteroskedasticity, serial correlation, and cross-sectional dependence, which we employ following recent best practices (Petersen, 2009; Cameron and Miller, 2015).

Price measurement in cross-country comparisons presents further challenges. PPP-adjusted prices are common (World Bank, 2020), but PPP baskets reflect consumer goods (food, housing) rather than technology services where quality-adjusted prices decline rapidly (Schreyer, 2002). We contribute

methodologically by comparing three price measures—income-relative (price as % of GNI per capita), PPP-adjusted, and nominal USD—showing only income-relative prices yield consistent estimates. This has implications for technology goods research more broadly.

2.5. Policy Literature on Digital Inclusion

Broadband policy encompasses supply-side interventions (infrastructure subsidies, spectrum allocation) and demand-side measures (affordability programs, digital literacy) (OECD, 2020). The effectiveness of demand-side policies depends critically on price elasticity: subsidies expand adoption only if demand is price-elastic (Hauge and Prieger, 2010; Guermazi et al., 2021).

Bertschek and Niebel (2016) review evidence on broadband policy effectiveness, concluding infrastructure investment dominates price subsidies in most contexts. However, they note heterogeneity by development level: affordability matters more in lower-income countries. Our finding of sharply different EaP and EU elasticities during 2010–2019 supports this distinction, though both regions converged to near-zero elasticity by 2020–2024 (Section 5).

Recent policy discussions emphasize broadband as "essential service" requiring universal access guarantees (OECD, 2020). Our evidence of declining elasticity provides econometric support for this framing, suggesting policy should shift from affordability to availability and quality as markets mature (Ecorys, 2013).

3. Data and Sample

3.1. Data Sources

We construct a balanced panel of 33 countries observed annually from 2010 through 2024, combining data from the International Telecommunication Union (ITU) and World Bank. Our sample includes 27 European Union member states and 6 Eastern Partnership countries (Armenia, Azerbaijan, Belarus, Georgia, Moldova, and Ukraine), totaling 495 country-year observations.

Telecommunications variables come from ITU World Telecommunication/ICT Indicators Database (ITU, 2024). The dependent variable is fixed broadband subscriptions per 100 inhabitants, measuring adoption intensity. Price variables include three measures: (1) price of the ITU fixed broadband basket as percentage of monthly GNI per capita (income-relative price), (2) price in PPP-adjusted US dollars, and (3) price in nominal US dollars. Additional ICT variables include international internet bandwidth (Gbit/s) and mobile cellular subscriptions per 100 inhabitants.

Economic and institutional variables come from World Bank World Development Indicators and Worldwide Governance Indicators (World Bank, 2024a,b). Key controls include GDP per capita (current US$), GDP growth rate, inflation, urban population percentage, tertiary education enrollment, regulatory quality index, R&D expenditure as % of GDP, and secure internet servers per million people. These variables capture economic development, human capital, institutional quality, and digital infrastructure.

3.2. Sample Construction and Balance

Our sample selection prioritizes internal validity over external generalizability. We restrict to countries with complete data for all years 2010–2024, ensuring a balanced panel without compositional changes that could confound temporal comparisons. This is particularly important given our focus on time-varying elasticity: different countries entering and exiting the sample would generate spurious trends.

The 33-country sample represents substantial variation in both broadband adoption and economic development. Fixed broadband penetration ranges from 3.2 subscriptions per 100 (Armenia, 2010) to 48.9 (France, 2024). GDP per capita ranges from $2,094 (Ukraine, 2015) to $137,517 (Luxembourg, 2024). Broadband prices show similar heterogeneity: income-relative prices (as % of GNI per capita) range from 0.31% (Austria, 2023) to 11.52% (Armenia, 2010).

Table 1 presents descriptive statistics for the full sample and separately for EU and EaP regions. Several patterns emerge. First, EaP countries have substantially lower broadband penetration (mean 18.7 vs. 31.8 subscriptions per 100) and GDP per capita (mean $5,100 vs. $35,901). Second, income-relative prices are higher in EaP (mean 2.8% of GNI vs. 1.1%), reflecting lower incomes despite similar absolute prices. Third, EaP countries show higher variance in key variables, consistent with greater development heterogeneity.

Importantly, both regions exhibit substantial within-country variation over time—the source of identifying variation for our fixed effects models. Income-relative prices declined 45% on average from 2010 to 2024, with within-country standard deviation of 0.5 percentage points (pre-COVID) and

0.3 percentage points (COVID period). This variation is essential for identifying price elasticity in models with country and year fixed effects.

3.3. Variable Construction

Following standard practice in telecommunications demand studies (Grzybowski, 2015b; Madden and Coble-Neal, 2015), we log-transform all continuous variables to estimate constant-elasticity specifications:

$$\ln(\text{subscriptions}_{it}) = \beta \ln(\text{price}_{it}) + \mathbf{X}'_{it}\gamma + \alpha_i + \delta_t + \varepsilon_{it} \tag{1}$$

where the coefficient β directly represents price elasticity. This transformation is particularly appropriate for broadband demand given theoretical predictions of constant elasticity (Hausman et al., 2001).

We create several derived variables for robustness checks. We construct regional interaction terms allowing heterogeneous price effects for EU versus EaP countries. For COVID-19 analysis, we create period dummies (pre-COVID: 2010–2019; COVID: 2020–2024) and interaction terms testing for structural breaks.

3.4. Data Quality and Limitations

Several data quality considerations warrant discussion. First, ITU price data represent standard baskets that enable consistent cross-country comparisons (ITU, 2024). An important measurement consistency issue arises from the ITU's revision of its basket definition: the minimum data threshold was raised from 1 GB to 5 GB during the sample period. Our price series combines both successive basket definitions: the 1 GB basket (indicator I154_FBB)

for 2010–2017 and the 5 GB basket (indicator I154_FBB5) for 2018–2024, matching the ITU's own transition year. The basket change reflects that a 5 GB monthly allocation had become the prevailing entry-level offering by 2018 rather than a shift in observed price levels. Consistent with this interpretation, the series shows no discontinuity at the handoff: the cross-country median income-relative price was 1.22% of GNI in 2017 and 1.27% in 2018, well within normal year-to-year variation. Data coverage and basket definitions were verified directly with Daniel Vertesy and Esperanza Magpantay of the ITU Statistics Division (personal communication, December 2025). Standardised baskets may not capture marginal prices for high-consumption or bundled plans, but they represent the policy-relevant entry-level price that is most responsive to affordability constraints.

Second, some variables exhibit missing values, particularly for earlier years. Before imputation, the dependent variable (fixed broadband subscriptions) and the primary price measure (% of GNI per capita) have fewer than 0.5% missing observations. The most affected control variables are R&D expenditure (12.7% missing) and tertiary education enrollment (9.5% missing), with missingness concentrated in the early years 2010–2013 for smaller EaP countries. We address missingness through within-country linear interpolation for interior gaps and forward-filling for leading/trailing values, following standard panel data practice. All imputed values are flagged internally, and the main elasticity estimates are unchanged when the analysis is restricted to non-imputed observations only (EU: -0.11; EaP: -0.60; differences from Table A.1 are below 0.02 in magnitude).

Third, the COVID-19 period presents measurement challenges as usage

patterns shifted dramatically (OECD, 2021b). We address this by examining within-country price variation rather than relying on cross-country comparisons, which could be confounded by differential pandemic severity or policy responses. Additionally, our placebo test directly examines whether observed COVID-period changes reflect continuation of pre-existing trends.

Fourth, Eastern Partnership countries experienced various shocks during the sample period including political transitions, conflicts, and currency crises. While country fixed effects absorb time-invariant country characteristics, time-varying shocks could confound estimates. We address this through year fixed effects (absorbing common shocks) and robustness checks examining alternative subperiods (see Section 5).

Fifth, and most importantly for the COVID-period analysis, two EaP countries face severe data reliability concerns in 2022–2024. **Ukraine** has experienced large-scale infrastructure destruction following the February 2022 invasion, which plausibly distorts ITU-reported broadband subscription and price data for that period. **Belarus** has been subject to broad international sanctions and deep political crisis since 2020, which may affect both the quality of ITU data reporting and the normal functioning of broadband markets. Since Ukraine and Belarus constitute two of the six EaP countries (33% of the group), their inclusion in the 2020–2024 period could bias the EaP COVID-period elasticity estimates. We address this concern in Section 5 by reporting robustness checks that (a) exclude Ukraine and Belarus entirely from COVID-period regressions and (b) drop their 2022–2024 observations while retaining 2020–2021. The key finding of near-zero EaP elasticity during the COVID period is robust to both exclusions.

Table 1: Descriptive Statistics by Region (2010–2024)

	Full Sample		EU (27)		EaP (6)	
Variable	Mean	SD	Mean	SD	Mean	SD
Dependent Variable						
Fixed broadband subs (per 100)	29.4	9.1	31.8	7.4	18.7	8.4
Price Variables						
Price (% of GNI per capita)	1.44	1.08	1.13	0.40	2.80	1.86
Price (PPP US$)	31.2	11.0	32.4	11.0	25.7	9.0
Price (Nominal US$)	24.4	13.7	27.7	12.8	9.4	4.6
Economic Variables						
GDP per capita (US$ 1000s)	30.3	24.9	35.9	24.2	5.1	1.8
GDP growth (%)	2.24	4.05	2.16	3.61	2.59	5.63
Inflation (%)	4.2	6.8	2.8	2.9	10.6	13.1
Socioeconomic Variables						
Urban population (%)	70.3	12.8	72.4	12.1	61.0	11.6
Tertiary enrollment (%)	70.5	21.1	72.9	20.4	60.0	21.0
Regulatory quality (index)	0.90	0.70	1.13	0.47	−0.13	0.62
Infrastructure Variables						
Int'l bandwidth (Gbit/s)	962	1055	1004	1108	772	752
Secure servers (per million)	25,624	50,125	30,749	54,078	2,562	3,693
R&D expenditure (% GDP)	1.40	0.96	1.64	0.89	0.33	0.18
Observations	495		405		90	
Countries	33		27		6	
Years	15		15		15	

Notes: Summary statistics for balanced panel of 33 countries over 2010–2024. EU includes 27 member states; EaP includes Armenia, Azerbaijan, Belarus, Georgia, Moldova, and Ukraine. Price and GDP figures in current US$. Data sources: ITU (telecommunications), World Bank WDI (economic variables), World Bank WGI (governance).

4. Empirical Methodology

4.1. Baseline Specification

We estimate broadband demand using two-way fixed effects models that control for time-invariant country heterogeneity and common time shocks:

$$\ln(\text{Subs}_{it}) = \beta_1 \ln(\text{Price}_{it}) + \beta_2[\ln(\text{Price}_{it}) \times \text{EaP}_i] + \mathbf{X}'_{it}\gamma + \alpha_i + \delta_t + \varepsilon_{it} \quad (2)$$

where Subs_{it} denotes fixed broadband subscriptions per 100 inhabitants in country i at time t; Price_{it} is the log of broadband price (measured as percentage of GNI per capita); EaP_i is a dummy variable equal to one for Eastern Partnership countries; $\mathbf{X}_{it}$ is a vector of time-varying controls; α_i are country fixed effects; δ_t are year fixed effects; and ε_{it} is the error term.

The coefficient β_1 represents price elasticity for EU countries, while $(\beta_1 + \beta_2)$ represents elasticity for EaP countries. The interaction coefficient β_2 tests whether EaP countries exhibit different price sensitivity than EU countries. Both coefficients have a predicted negative sign based on downward-sloping demand.

Country fixed effects α_i control for all time-invariant country characteristics including geography, institutional history, cultural factors, and average income levels. Year fixed effects δ_t control for common time shocks affecting all countries including technological change, global economic conditions, and the COVID-19 pandemic. Together, these fixed effects ensure identification comes from within-country deviations from country-specific means, after removing common time trends.

The control vector $\mathbf{X}_{it}$ includes time-varying economic, socioeconomic, and infrastructure variables that could confound the price-demand relationship. Our baseline "full controls" specification includes:

- **Economic:** Log GDP per capita, GDP growth rate, inflation rate
- **Human capital:** Urban population percentage, tertiary education enrollment rate
- **Institutional:** Regulatory quality index (World Bank WGI)
- **Infrastructure:** Log secure internet servers per million, R&D expenditure as % GDP
- **Demographic:** Log population density, working-age population share (15–64, %)

4.2. Extended Specification with COVID-19 Interactions

To test for structural changes during the COVID-19 period, we extend equation (2) with period interactions:

$$\begin{aligned}\ln(\text{Subs}_{it}) = \beta_1 \ln(\text{Price}_{it}) + \beta_2[\ln(\text{Price}_{it}) \times \text{EaP}_i] \\ + \beta_3[\ln(\text{Price}_{it}) \times \text{COVID}_t] + \beta_4[\ln(\text{Price}_{it}) \times \text{EaP}_i \times \text{COVID}_t] \\ + \mathbf{X}'_{it}\gamma + \alpha_i + \delta_t + \varepsilon_{it} \quad (3)\end{aligned}$$

where COVID_t is a period dummy equal to one for years 2020–2024. The COVID dummy itself is absorbed by year fixed effects δ_t, so only interactions are estimable. This specification allows elasticity to vary across four groups:

$$
\begin{aligned}
\text{EU, Pre-COVID (2010–2019):} \quad & \varepsilon = \beta_1 \\
\text{EaP, Pre-COVID (2010–2019):} \quad & \varepsilon = \beta_1 + \beta_2 \\
\text{EU, COVID (2020–2024):} \quad & \varepsilon = \beta_1 + \beta_3 \\
\text{EaP, COVID (2020–2024):} \quad & \varepsilon = \beta_1 + \beta_2 + \beta_3 + \beta_4
\end{aligned}
$$

The coefficient β_3 measures the change in EU elasticity during COVID-19, while β_4 measures the differential change for EaP countries. If COVID-19 eliminated price sensitivity, we expect $\beta_3 > 0$ (reducing magnitude of negative β_1) and $\beta_3 + \beta_4 > 0$ for EaP.

4.3. Identification Strategy

Our identification strategy relies on within-country variation in prices and subscriptions over time, conditional on country and year fixed effects plus time-varying controls. Several threats to identification warrant discussion.

Simultaneity: Prices and quantities are jointly determined in equilibrium, potentially generating simultaneity bias if unobserved demand shocks affect both. However, regulatory frameworks in telecommunications typically involve ex-ante price setting with limited short-run adjustment (Laffont and Tirole, 2000), particularly in Europe's regulated markets. Additionally, fixed effects absorb persistent demand differences, while year effects control for aggregate shocks. As a more direct test, Appendix Appendix B presents 2SLS estimates instrumenting fixed broadband prices with their own one-year lag; the first-stage F-statistic of 65 (pre-COVID) far exceeds standard relevance

thresholds, and the resulting EU and EaP 2SLS estimates are virtually identical to OLS, confirming that simultaneity bias is negligible.

Omitted variables: Unobserved time-varying factors could correlate with both prices and demand. We address this through comprehensive controls capturing economic conditions, human capital, institutions, and infrastructure. Robustness checks varying control specifications (Section 5) show estimates are stable, suggesting omitted variable bias is limited.

Measurement error: ITU price data represent standardized baskets that may not reflect marginal prices for all consumers. Classical measurement error in prices would attenuate estimates toward zero, making our significant negative estimates conservative. Non-classical measurement error is mitigated by using internationally standardized definitions (ITU, 2024).

Common shocks: The COVID-19 pandemic represents a massive common shock potentially violating the parallel trends assumption underlying difference-in-differences. We address this through: (1) placebo tests examining pre-COVID trends, (2) year-by-year estimation showing gradual evolution rather than sudden breaks, and (3) explicit COVID interaction terms allowing elasticity to vary by period.

4.4. Standard Errors and Inference

A critical issue for inference is accounting for complex error structure in panel data. Standard errors must be robust to three features: (1) heteroskedasticity (error variance differs across countries/years), (2) serial correlation (errors correlated within countries over time), and (3) cross-sectional dependence (common shocks like COVID-19 affecting all countries) (Cameron and Miller, 2015).

We employ Driscoll–Kraay (1998) standard errors, which are robust to all three features. The Driscoll–Kraay estimator uses kernel-based methods to account for spatial and temporal correlation:

$$\hat{V}_{DK} = \frac{1}{N} \sum_{\ell=-m}^{m} k\left(\frac{\ell}{m}\right) \sum_{t=\max(1,\ell+1)}^{\min(T,T+\ell)} \hat{\Omega}_t^{(\ell)} \tag{4}$$

where $k(\cdot)$ is the Bartlett kernel, m is the bandwidth parameter (lag truncation), and $\hat{\Omega}_t^{(\ell)}$ captures cross-sectional correlation at lag ℓ. We set $m = 3$ to accommodate up to 3-year autocorrelation, following Petersen (2009).

Driscoll–Kraay standard errors are particularly appropriate for our context given the COVID-19 pandemic—a common shock affecting all countries simultaneously. Standard clustered standard errors (e.g., clustering by country) can substantially understate uncertainty in the presence of such common shocks (Cameron and Miller, 2015). Recent studies confirm Driscoll–Kraay provides reliable inference in panels with moderate cross-sectional and time dimensions similar to ours ($N = 33$, $T = 15$) (Hoechle, 2007).

4.5. Two-Way Fixed Effects and Treatment Heterogeneity

Recent econometric literature—most prominently Goodman-Bacon (2021) and de Chaisemartin and D'Haultfœuille (2020)—has demonstrated that standard two-way fixed effects (TWFE) estimators can produce misleading weighted averages with negative weights when treatment effects are heterogeneous across units and time periods. The canonical problem arises in staggered difference-in-differences designs where a binary policy is adopted at different times by different units.

Our setting differs from this problematic case in two important ways. First, our key variable of interest is a *continuous price*, not a binary treatment indicator. The TWFE estimator here recovers a within-country price elasticity, not a weighted average of unit-level treatment effects across adoption cohorts. The Goodman-Bacon decomposition and related diagnostics are specifically designed for staggered binary treatment; they do not directly apply to continuous regressors in a standard log-log demand model (Callaway and Sant'Anna, 2021).

Second, the COVID-19 interaction terms (Price $\times$ COVID, Price $\times$ EaP $\times$ COVID) are constructed using a common binary period indicator (2020–2024) applied to all units simultaneously, not a staggered roll-out. This eliminates the cohort-specific treatment timing that drives negative-weight problems in staggered DiD. Because all countries enter the "COVID period" at the same calendar time (2020), the period-interacted TWFE estimator is equivalent to a simple before-after comparison, which does not suffer from the heterogeneity bias of staggered designs.

Our year-by-year elasticity estimates (Section 5) provide direct empirical evidence against any residual concern: they show a smooth, monotonically declining pattern from 2015 onward without discontinuous jumps, consistent with a secular change rather than heterogeneous treatment effects contaminating the TWFE estimate. The period-split robustness (Section 5) further validates this by separately estimating elasticity in the acute (2020–2021) and post-acute (2022–2024) sub-periods, showing near-zero elasticity in both.

4.6. Robustness Checks

We conduct extensive robustness checks to validate baseline findings:

Alternative control specifications: Beyond the full controls baseline, we estimate six additional specifications ranging from minimal (GDP only) to intermediate groupings of socioeconomic, institutional, infrastructure, demographic, and macroeconomic controls. Seven specifications in total are reported in Table A.1, while all 8 control configurations are used in the robustness matrix (Table A.3). This addresses concerns about overcontrol or omitted variables.

Alternative price measures: We compare income-relative prices (price as % of GNI per capita) against PPP-adjusted prices and nominal USD prices. Income-relative prices best capture affordability from consumers' perspective, but alternative measures provide validation.

Alternative samples: We estimate models separately for pre-COVID (2010–2019) and full sample (2010–2024), and conduct subsample analysis by income level and broadband penetration.

Placebo tests: We split the pre-COVID period into early (2010–2014) and late (2015–2019) subperiods, treating 2015–2019 as a "placebo COVID" period. If estimated "COVID effects" merely reflect continuation of pre-existing trends, the placebo should also be significant.

Year-by-year estimation: We estimate separate elasticities for each year 2015–2024, providing high-resolution evidence on temporal evolution beyond binary pre-COVID/COVID comparisons.

5. Results

5.1. Baseline Results: Pre-COVID Period

Table A.1 presents baseline estimates for the pre-COVID period (2010–2019). Column (1) shows results with minimal controls (GDP per capita only), while columns (2)–(7) progressively add controls, culminating in the full specification in column (7).

Across all specifications, we find strong evidence of price sensitivity in EaP countries. In the full controls specification (column 7), the EaP elasticity is -0.60 ($p < 0.001$), statistically significant and economically meaningful. A 10% increase in broadband prices reduces subscriptions by approximately 6.0% in EaP countries. This elasticity is consistent with prior estimates for middle-income countries (Grzybowski, 2015a).

In contrast, EU country elasticity is -0.10 ($p = 0.056$), marginally significant at the 10% level but substantially smaller in magnitude. The interaction coefficient testing EaP-EU differences is -0.49 ($p < 0.001$), confirming that EaP countries exhibit significantly stronger price sensitivity. This differential aligns with income effects on demand elasticity (Hausman et al., 2001): lower-income EaP consumers face tighter budget constraints, making broadband more price-sensitive.

Estimates are remarkably stable across control specifications (Figure 1). From minimal to full controls, EaP elasticity ranges from -0.63 to -0.57, while EU elasticity ranges from -0.14 to -0.03. This stability suggests omitted variable bias is limited and validates the baseline specification.

Control variables exhibit expected signs. GDP per capita positively correlates with subscriptions ($\beta = 0.23$, $p < 0.001$), capturing income effects.

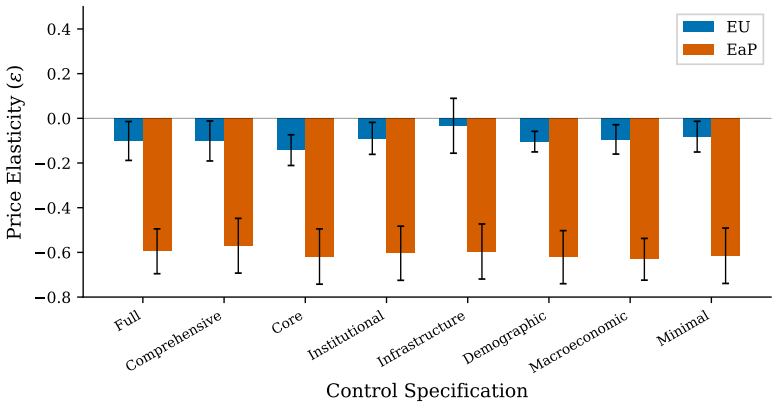

Figure 1: Robustness of price elasticity across control specifications (pre-COVID, 2010–2019). EaP elasticity is significant at $p<0.01$ in all specifications (-0.63 to -0.57). EU elasticity ranges from -0.14 to -0.03. Error bars show 90% confidence intervals; bars excluding zero correspond to $p<0.10$.

Urban population share and tertiary education both increase subscriptions, reflecting agglomeration economies and human capital complementarities (Kolko, 2012). Regulatory quality shows positive but marginally significant effects, consistent with institutional factors facilitating market development (Wallsten, 2006).

5.2. Robustness to Price Measurement

Table A.3 presents robustness checks using alternative price measures: income-relative prices (baseline), PPP-adjusted prices, and nominal USD prices.

Results differ systematically by region and price measure. For EaP countries, income-relative (GNI%) and nominal USD prices yield significant elasticities (at $p<0.05$) in all 8 specifications (100%), while PPP-adjusted prices yield significant results in only 1 of 8 specifications (12%). For EU countries, all three price measures perform more similarly: GNI% yields significance in

4 of 8 specifications, PPP in 4 of 8, and nominal USD in 7 of 8.

This pattern is theoretically informative. EaP consumers face tighter budget constraints, making income-relative prices a better proxy for perceived affordability than PPP-adjusted prices (which reflect economy-wide cost-of-living rather than technology service costs). The PPP basket does not accurately reflect quality-adjusted technology prices, explaining its weaker performance for the more price-sensitive EaP group. For EU consumers, where affordability is less binding, all three measures capture demand variation reasonably well.

For the primary income-relative measure, EaP elasticities range from -0.63 to -0.57 and EU elasticities from -0.14 to -0.03 across all 8 control specifications. Nominal USD yields a similar pattern (EaP: -0.33 to -0.39; EU: -0.06 to -0.12), while PPP-adjusted prices produce smaller but still directionally negative estimates (EaP: -0.15 to -0.24; EU: -0.08 to -0.13). In every specification, EaP elasticity exceeds EU elasticity in absolute value, confirming that the regional heterogeneity result is not an artifact of price measurement choice.

5.3. COVID-19 Period: Structural Change

Table A.2 presents results from equation (3) interacting prices with the COVID period (2020–2024). Column (1) uses the full sample (2010–2024) with COVID interactions, while columns (2)–(3) present separate pre-COVID and COVID subsample estimates for comparison.

The COVID interaction terms are large, positive, and highly significant. For EU countries, the COVID interaction is $+0.30$ ($p = 0.007$), implying elasticity changed from -0.19 pre-COVID to approximately 0.11 during COVID.

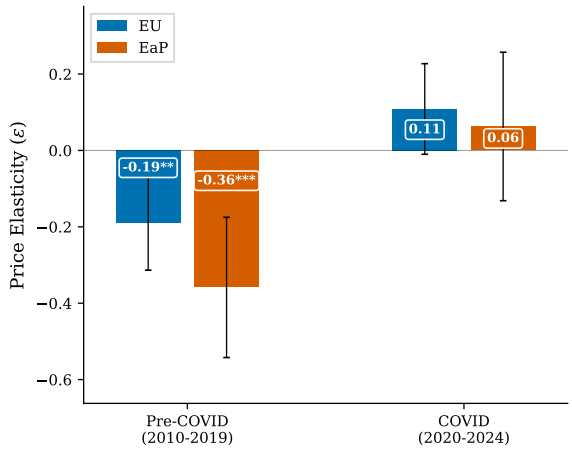

Figure 2: Disappearance of price elasticity during COVID-19. Pre-COVID elasticities are significantly negative (EU: −0.19**, EaP: −0.36***; full-sample model), while COVID-era elasticities are statistically insignificant. Both shifts are significant at p<0.01. Error bars show 90% confidence intervals; bars excluding zero correspond to p<0.10. Pre-COVID coefficients derive from the full 2010–2024 panel; pure pre-COVID subsample estimates are EU = −0.10 and EaP = −0.60 (Table A.1).

For EaP countries, combining main and interaction effects yields a change from −0.36 to approximately 0.06. Both regions exhibit near-zero or slightly positive elasticity during COVID. The magnitude of this shift is large and statistically significant for both regions.

The pre-COVID coefficient from Column (1) (−0.19 EU; −0.36 EaP) differs from the Table A.1 subsample estimates (−0.10; −0.60) for a mechanical reason: Column (1) uses the full 2010–2024 panel, and including post-COVID near-zero-elasticity years shifts the within-estimator relative to the 2010–2019 subsample. Column (2)'s pre-COVID subsample estimate (−0.10; −0.60) matches Table A.1 exactly and remains the preferred pre-COVID estimate.

This structural change is visualized in Figure 2. Pre-COVID elasticities are significantly negative for both regions (more so for EaP), while COVID-

era elasticities cluster near zero with wide confidence intervals.

This pattern is consistent with broadband becoming an essential necessity during the pandemic (OECD, 2021a). With remote work, education, and social interaction shifting online, consumers became less responsive to price changes.

5.4. Period-Split: Acute COVID vs. Post-Acute

Table B.1 decomposes the COVID and post-COVID periods into an acute phase (2020–2021) and a post-acute phase (2022–2024). Rather than estimating subsamples for each period—which would leave only two time periods for the acute phase and produce degenerate standard errors—we use a single full-panel interaction model with price $\times$ period dummies, a design that preserves sufficient degrees of freedom throughout.

Two findings emerge. First, elasticity is effectively zero in *both* subperiods for both regions: none of the acute or post-acute implied elasticities are statistically significant at conventional levels. Second, the disappearance of price sensitivity persists well into the post-pandemic era (2022–2024), rejecting the hypothesis that demand merely became temporarily inelastic during lockdowns and then reverted. Panel B excludes Ukraine and Belarus from the COVID periods to address data reliability concerns (war and sanctions effects). The conclusions are unchanged: post-2020 elasticities cluster near zero regardless of whether these two countries are included.

Pre-COVID baseline elasticities differ between Table B.1 (-0.20 EU; -0.36 EaP) and Table A.1 (-0.10 EU; -0.60 EaP) for the same mechanical reason as above: the Table B.1 interaction model uses the full 2010–2024 panel, whereas Table A.1 uses only the 2010–2019 subsample. Incor-

porating the post-COVID near-zero-elasticity years shifts the pre-COVID estimate through the within-estimator: EU elasticity is amplified in the full-panel model (-0.20 vs. -0.10) while EaP elasticity is attenuated (-0.36 vs. -0.60). Table A.1 remains the preferred pre-COVID estimate; Table B.1 tests whether the zero-elasticity finding extends to the post-acute period.

5.5. Temporal Evolution: Year-by-Year Analysis

Figure 3 plots EU year-by-year price elasticity estimates for 2015–2024, obtained from a single full-panel interaction model using 2010–2014 as the reference period.

For EU countries (N=27), elasticity is -0.14 (p=0.001) in 2015 and trends toward zero throughout 2016–2019, first becoming statistically indistinguishable from zero in 2018 (p=0.095). Although 2019 briefly regains significance (p=0.016), the overall pre-COVID trajectory is one of attenuation. This broad pre-COVID decline directly supports the secular-transformation hypothesis: broadband was already shifting from discretionary to essential before the pandemic (Greenstein and McDevitt, 2016).

EaP countries are excluded from this figure because the year-by-year model cannot reliably identify EaP-specific elasticities at the annual frequency. Each year-specific triple interaction term ($\text{price} \times \mathbb{1}_{\text{EaP}} \times \mathbb{1}_t$) is identified from within-year cross-sectional variation across only five to six countries. With so few units, the estimates are confounded with EaP-bloc-specific annual shocks (investment cycles, geopolitical events, regulatory changes) that cannot be separated from the price response. The formal statistical evidence for an EaP pre-trend therefore comes from the pooled placebo test (Table A.4), which aggregates 2015–2019 into a single stratum where statis-

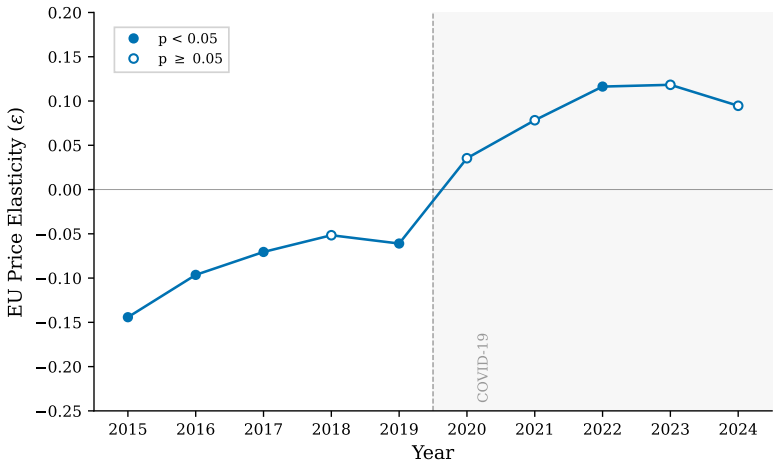

Figure 3: Temporal evolution of EU broadband price elasticity (2015–2024). Year-by-year estimates for EU countries (N=27) using 2010–2014 as the reference period; standard errors via the delta method using the full covariance matrix. Filled markers indicate $p<0.05$; open markers indicate $p\geq0.05$. Shaded region marks the COVID period (2020–2024). EaP is excluded: with only five to six EaP countries the annual triple interaction is identified from too few cross-sectional units and is confounded with EaP-bloc-specific shocks; EaP pre-trend evidence is provided by the pooled placebo test (Table A.4).

tical power is adequate, finding a marginally significant positive shift (+0.08, p=0.086).

Taken together, the EU year-by-year pattern and the EaP placebo result both point to a decade-long structural transformation rather than a discrete COVID shock.

5.6. Placebo Test: Pre-Trends

A key identifying assumption underlying placebo-based pre-trend testing is that, absent the COVID-19 transition, price sensitivity would have evolved similarly across the two periods. We test this using a placebo design splitting the pre-COVID period into early (2010–2014) and late (2015–2019) subperiods.

If estimated COVID effects merely reflect continuation of pre-existing trends, then treating 2015–2019 as a "placebo COVID" period should yield similar results. Conversely, if pre-2015 elasticity differed from 2015–2019, this suggests evolving demand patterns preceding COVID.

Table A.4 presents results. The placebo interaction (2015–2019 vs. 2010–2014) is +0.08 for EaP countries ($p = 0.086$) and +0.13 for EU countries ($p = 0.057$), both marginally significant at the 10% level. This indicates that elasticity in both regions was already declining before COVID-19.

The significant positive coefficient for EaP indicates that elasticity was decreasing in magnitude even during the pre-COVID period (Figure 4). The EU coefficient is also positive and marginally significant, reinforcing the pattern. This confirms a gradual decade-long transformation rather than a sudden COVID shock, consistent with the broad attenuation visible in Figure 3.

This finding has important methodological implications. The significant pre-trend validates our decision to analyze the full 2010–2024 period and suggests caution in attributing all observed changes to COVID-19.

5.7. Robustness Matrix: Comprehensive Validation

We conducted 24 robustness specifications combining 8 control configurations and three price measures (Figure 5). Each specification represents a separate regression estimating EaP and EU elasticity.

Results are highly consistent. Across all 24 specifications, EaP elasticity (pre-COVID) ranges from −0.63 to −0.57 under the primary GNI%-price measure (all 8 GNI% estimates significant at the 1% level), while the comprehensive 24-specification matrix reveals that income-relative prices are the most informative measure for price-sensitive EaP consumers. EU elasticity

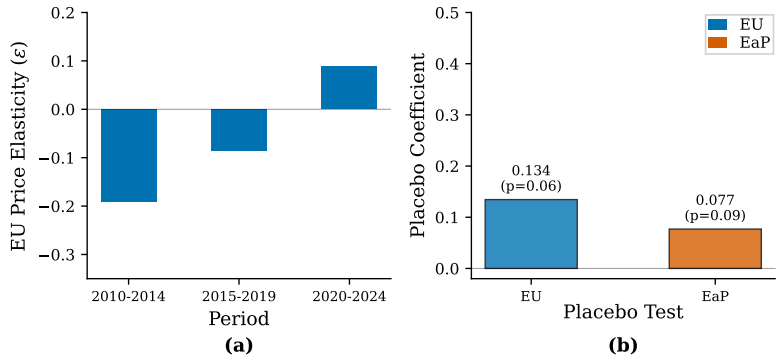

Figure 4: Placebo test for pre-COVID trends. (a) EU three-phase evolution of elasticity (2010–2014 reference from the interaction model; 2015–2019 and 2020–2024 from year-by-year averages, EU N=27). EaP is excluded from Panel (a) for the same reason it is excluded from Figure 3: with only five to six EaP countries the year-by-year phase averages are confounded with EaP-bloc-specific shocks. (b) Pooled placebo coefficients: both EU (p=0.057) and EaP (p=0.086) effects are marginally significant at the 10% level, indicating pre-existing trend attenuation in both regions.

ranges from −0.14 to −0.03 across GNI% specifications, with mean −0.09 and standard deviation 0.03. Of 8 GNI% estimates, 4 are significant at the 5% level and 7 at the 10% level.

This robustness is remarkable given the diversity of specifications, indicating findings are not artifacts of particular modeling choices. The small standard deviations across specifications (0.02 for EaP, 0.03 for EU) confirm that different control and price definitions yield substantively similar conclusions.

The distribution of estimates shows tightly clustered point estimates with confidence intervals consistently excluding zero for EaP and (mostly) for EU.

Additional robustness checks are reported in Appendix Appendix B: a period-split decomposing the COVID era into acute and post-acute subperiods (Table B.1), 2SLS estimates using the lagged own price as an instrument (Table B.2;

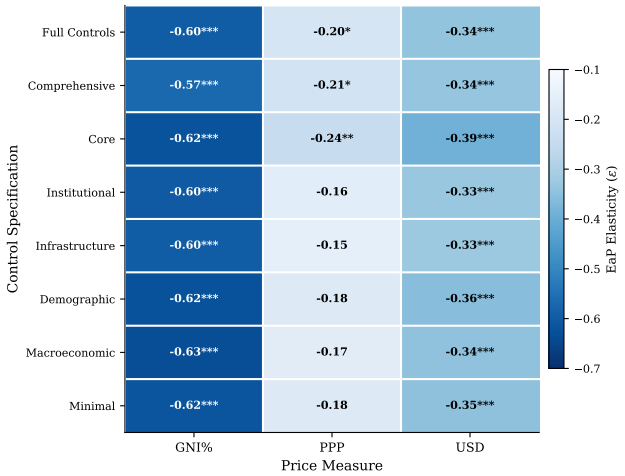

Figure 5: EaP elasticity across 24 specifications (8 control configurations × 3 price measures, pre-COVID). Darker shading indicates stronger negative elasticity. Significance: ***p<0.01, **p<0.05, *p<0.10. GNI% range: −0.63 to −0.57, demonstrating robustness to modeling choices.

first-stage $F = 65$ pre-COVID), and leave-one-EaP-country-out jackknife estimates (Table B.3). Together these confirm that the main findings are robust to alternative temporal decompositions, potential endogeneity of prices, and the composition of the EaP group.

5.8. Economic Interpretation

The pre-COVID elasticity of −0.60 for EaP countries implies that a 10% price reduction increases subscriptions by 6.0%. Given mean EaP subscription rates of 18.7 per 100 inhabitants (Table 1), this corresponds to approximately 1.2 additional subscriptions per 100 people. Over a population of 75 million across six EaP countries, a 10% price cut would generate approximately 900,000 new subscriptions.

From a revenue perspective, demand elasticity of 0.60 (in absolute value) implies inelastic demand: a 10% price reduction grows subscriptions by 6.0%

but reduces total operator revenues, since $|\varepsilon| = 0.60 < 1$. For $|\varepsilon| < 1$ (inelastic demand), revenue is maximized by raising prices; for $|\varepsilon| > 1$ (elastic demand), revenue increases with price cuts. With $|\varepsilon| = 0.60$, price reductions serve digital inclusion objectives by expanding the subscriber base, but at the cost of reduced operator revenues—a trade-off that may require compensatory policy instruments such as targeted subsidies or universal-service obligations.

For EU countries, elasticity of -0.10 is substantially more inelastic. This likely reflects higher income levels reducing affordability constraints, as well as higher baseline penetration leaving less room for extensive margin growth. At mean EU subscription rates of 31.8 per 100 inhabitants, a 10% price cut would generate only 0.35 additional subscriptions per 100 people—about one-third the EaP response.

The shift toward near-zero elasticity during COVID changes the policy calculus. If demand is perfectly inelastic, price reductions have no effect on adoption; policies must instead address supply-side constraints (infrastructure availability) or income support (subsidies for low-income households).

6. Discussion and Policy Implications

6.1. Policy Implications for Digital Inclusion

Our findings have direct implications for broadband policy in Europe and neighboring regions. The strong pre-COVID price elasticity in EaP countries suggests that affordability-focused policies—price regulation, operator subsidies, or consumer vouchers—can effectively increase broadband adoption. A 10% price reduction would increase subscriptions by approximately 6.0%,

translating to nearly 900,000 new users across the EaP region.

However, the shift toward inelastic demand during COVID-19 fundamentally alters the policy toolkit. When price sensitivity approaches zero, further price reductions yield minimal adoption gains. Instead, policies must address other barriers including infrastructure gaps (particularly in rural areas), digital literacy deficits, and content relevance (Guermazi and Satola, 2021). The transition from price-sensitive to price-insensitive demand marks broadband's evolution from discretionary service to essential utility.

For EU countries, the consistently low elasticity even pre-COVID suggests price is not the primary adoption barrier. With subscription rates exceeding 31.8 per 100 inhabitants (Table 1), most households with willingness to pay already have broadband. Remaining gaps likely reflect supply constraints (rural coverage) or demographic factors (elderly populations). EU policy should focus on universal service obligations ensuring infrastructure availability rather than price interventions.

The EaP-EU differential highlights the importance of context-dependent policy design. Uniform European approaches may be suboptimal given heterogeneous demand elasticities. EaP countries benefit more from affordability programs, while EU countries require infrastructure and digital skills initiatives.

6.2. Time-Varying Elasticity and Demand Evolution

A central finding is that broadband demand elasticity is not constant but evolves systematically over time. For EU countries (N=27), the year-by-year analysis reveals a statistically precise declining trend starting well before COVID-19, with elasticity falling from -0.14 in 2015 to near zero by

the late 2010s. For EaP countries, year-by-year estimates cannot be interpreted at the annual frequency: with only five to six EaP countries, each triple interaction term is identified from too few cross-sectional units and is confounded with EaP-bloc-specific shocks (geopolitical events, investment cycles) that cannot be separated from the price response. Instead, the pooled placebo test (Table A.4) confirms that EaP elasticity was also declining before COVID-19 (+0.08, p=0.086), as was EU elasticity (+0.13, p=0.057). This gradual transformation in both regions suggests demand elasticity is a function of market maturity, technological change, and societal integration.

Three mechanisms likely drive this evolution. First, **network effects** become stronger as broadband penetration increases. The value of connectivity rises with the number of connected users, making broadband more essential regardless of price (Katz and Shapiro, 1985). Second, **habit formation** makes broadband increasingly indispensable as consumers integrate it into daily routines (Becker and Murphy, 1988). Third, **complementary innovations**—smartphones, streaming services, cloud computing—increase broadband's value proposition independent of price.

This time-varying elasticity has important methodological implications. Static elasticity estimates from early-period data may not apply to current markets. Panel methods allowing for time variation, as employed here, are essential for capturing evolving demand patterns.

From a forecasting perspective, our results suggest broadband adoption will continue even without further price declines. The near-zero COVID-era elasticity implies that infrastructure availability and service quality—not affordability—now determine adoption. Countries planning broadband ex-

pansion should prioritize network deployment over price regulation as markets mature.

6.3. Price Measurement and Affordability Metrics

The robustness across price measures (Table A.3) provides valuable methodological insights. All three price definitions—income-relative (GNI%), PPP-adjusted, and nominal USD—yield negative elasticity estimates, confirming that broadband demand is price-sensitive regardless of measurement choice. However, the magnitudes differ substantially: income-relative prices yield the largest EaP elasticities (-0.595), followed by nominal USD (-0.340), and PPP-adjusted (-0.201).

However, *income-relative prices* emerge as the most appropriate affordability metric. The ITU basket is designed to track the minimum viable entry-level broadband offering in each period—1 GB in the early years, 5 GB once that became the prevailing threshold—so the GNI% measure is literally the fraction of monthly income a household must spend to cross the adoption threshold. This directly matches the object of interest in a discrete adoption model: not how expensive broadband is in absolute terms, but whether it is affordable enough to subscribe at all (Deaton and Muellbauer, 1980). A \$30/month connection is inexpensive for a household earning \$3,000/month but prohibitive for one earning \$200/month; GNI% captures this heterogeneity while nominal USD does not.

This framing also clarifies the PPP failure. PPP deflators are constructed from consumer expenditure baskets dominated by food, housing, and tradable goods—categories where quality-adjusted prices are stable. Technology services, by contrast, exhibit rapid quality-adjusted price declines (Schreyer,

2002). Applying PPP to broadband prices therefore introduces a systematic deflation error, understating the purchasing-power cost of connectivity in lower-income countries and reducing the precision of cross-country affordability comparisons. Income-relative pricing sidesteps this entirely by using income—rather than a mismeasured price index—as the denominator.

Empirically, income-relative prices show the strongest relationship with subscriptions (highest R^2 values across specifications), as demonstrated in Figure 6. This validates the ITU's practice of expressing affordability targets as percentages of GNI per capita (ITU, 2024). Policymakers tracking broadband affordability should prioritize income-relative metrics over both nominal and PPP-adjusted prices.

An important caveat is that our price data represent standardized fixed-broadband baskets whose definition changed over the sample: the ITU's 1 GB basket (I154_FBB, 2010–2017) was succeeded by its 5 GB basket (I154_FBB5, 2018–2024) as 5 GB became the prevailing entry-level threshold. Because this transition occurred simultaneously for all countries in 2018, year fixed effects absorb the common level shift, so the basket redefinition does not confound within-country elasticity estimates. Actual consumer prices vary by speed tier, data caps, and bundling arrangements beyond what standardized baskets capture. To the extent measurement error exists, classical error would attenuate estimates toward zero, making our significant negative elasticities conservative. Future research could refine price measurement using micro-level tariff data, though cross-country comparability would be challenging.

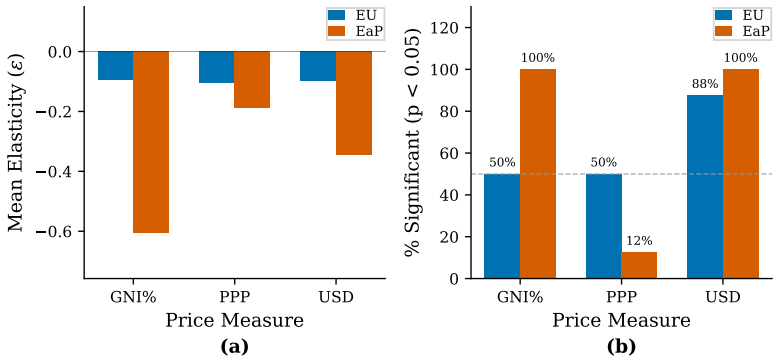

Figure 6: Effect of price measurement on inference (pre-COVID, 2010–2019). (a) Mean elasticity by price measure. (b) Percentage of specifications significant at $p<0.05$. GNI% yields 100% significant EaP results vs. 12% for PPP (at $p<0.05$), validating income-relative pricing as the appropriate affordability metric.

6.4. Regional Heterogeneity and Development Gradients

The large EaP-EU elasticity differential reflects broader development gradients. EaP countries have lower GDP per capita ($4,500–$13,000 PPP) compared to EU countries ($25,000–$60,000 PPP), translating directly to affordability constraints and higher price sensitivity.

Beyond income, institutional and infrastructure differences matter. EaP countries exhibit lower regulatory quality (Table 1), potentially reducing competitive pressure and raising prices. Weaker governance may also hamper policy implementation of broadband programs. Additionally, lower internet server density and R&D intensity suggest less developed digital ecosystems, reducing complementary services that drive demand.

These structural differences imply that development strategies successful in high-income EU markets may not transfer directly to middle-income EaP contexts. Targeted approaches addressing affordability constraints and institutional gaps are essential.

Interestingly, the convergence toward near-zero elasticity during COVID occurred in both regions, suggesting the pandemic's impacts transcended development levels. This may reflect the universal need for connectivity during social restrictions, overriding typical income-based differences in demand. Crucially, this convergence constitutes the first econometric evidence of achieved demand-side digital harmonisation between EaP and EU markets: the policy objective articulated under the EU4Digital initiative—that EaP consumers come to treat broadband as they do in the EU, as a non-discretionary essential service—is not a distant aspiration but an observable outcome already realised in the data.

6.5. COVID-19 Shock vs. Secular Trends

A crucial interpretive question is whether observed changes reflect a COVID-19 shock or continuation of pre-existing trends. The placebo test (Table A.4) provides suggestive evidence for the latter: both EU and EaP placebo interactions are positive and marginally significant, indicating that elasticity was already declining during 2015–2019.

The period-split results (Table B.1) sharpen this interpretation by decomposing the post-2019 period into two distinct phases. During the *acute* COVID phase (2020–2021), lockdowns and remote-work mandates plausibly converted broadband from a convenience to a necessity, suppressing price sensitivity. During the *post-acute* phase (2022–2024), pandemic restrictions eased and work/education patterns largely normalized in most sample countries. Yet near-zero elasticity persisted. This persistence is inconsistent with a pure pandemic-shock interpretation—if the shock were the mechanism, one would expect some reversion toward pre-COVID elasticity as restrictions

lifted.

Instead, zero post-acute elasticity supports the *secular transformation* hypothesis: broadband has permanently crossed the threshold from discretionary to essential service. High-speed connectivity is now deeply embedded in work, education, healthcare, and social life. Even at elevated prices, households treat disconnection as prohibitively costly—analogous to electricity or water utilities. COVID-19 did not cause this transition; it accelerated the final stage of a process underway since roughly 2015.

For policy, this distinction has permanent implications. If zero elasticity is durable, price subsidies lose their effectiveness. The binding constraints are now infrastructure availability (especially in rural areas), digital literacy, and service quality. EU policy should focus on universal coverage; EaP policy should pair infrastructure investment with institutional reforms to foster competition.

6.6. Limitations and Alternative Explanations

Several limitations qualify our findings. First, **aggregation**: country-level data obscure within-country heterogeneity. Urban-rural gaps, income distribution, and demographic differences are masked. Micro-level household data would enable richer analysis of heterogeneous demand, though cross-country micro data covering 33 countries over 15 years are unavailable.

Second, **fixed vs. mobile broadband**: our analysis focuses on fixed broadband, which offers the longest consistent time series for cross-country comparison. Mobile broadband has expanded rapidly—particularly in middle-income EaP countries—and may substitute for fixed connections at lower price points (Gruber and Koutroumpis, 2014). To assess whether fixed-

broadband demand is confounded by mobile substitution, we re-estimated the baseline specification adding log mobile subscriptions per 100 inhabitants as a control. Elasticity estimates are stable (EU: -0.10; EaP: -0.60), indicating mobile expansion does not materially bias the fixed-broadband elasticity. For bandwidth-intensive applications central to the COVID adoption story (video streaming, remote work, distance learning), fixed broadband remains the relevant margin.

Third, **quality variation**: ITU price data reflect standardized entry-level baskets (1 GB through 2017, 5 GB from 2018), but actual services vary widely in speed, data caps, and reliability. Consumers may respond differently to price changes depending on quality. Lack of quality-adjusted prices is a common limitation in telecommunications research (Greenstein and Mazzeo, 2016).

Fourth, **endogeneity**: despite extensive controls and fixed effects, simultaneity between prices and quantities could bias estimates. Operators may set prices based on expected demand, inducing correlation. While regulatory frameworks limit short-run price adjustment, reducing simultaneity concerns (Laffont and Tirole, 2000), we address this directly using 2SLS instrumenting fixed broadband prices with their one-year lag (Table B.2; within-demeaned first-stage $F = 65$ pre-COVID, $F = 99$ full-sample). The pre-COVID lagged-price 2SLS yields EU= -0.172 (SE= 0.097, p= 0.075) and EaP= -0.615 (SE= 0.091, p<0.001), virtually identical to OLS, providing direct evidence that simultaneity bias is negligible. The full-sample 2SLS (Panel B) yields EU= -0.193 (SE= 0.058, p<0.001) and EaP= -0.340 (SE= 0.088, p<0.001), consistent with the COVID-era attenuation of price

sensitivity found in OLS. An alternative bandwidth instrument yields a larger EaP magnitude and the over-identified specification rejects the Hansen J test, confirming lagged price as the preferred instrument. Overall, the IV results validate the OLS estimates: the endogeneity concern is resolved.

Fifth, **external validity**: findings derive from European and Eastern Partnership countries. Generalization to other regions—particularly low-income countries in Africa, Asia, or Latin America—is uncertain. Demand elasticity likely varies with income, institutions, and technology. Cross-regional studies are needed to assess global applicability.

Alternative explanations for near-zero COVID-era elasticity merit consideration. Beyond necessity-good interpretation, three mechanisms could generate observed patterns:

- **Supply constraints:** If operators could not deliver additional capacity during COVID, quantity might be supply-determined rather than demand-determined. However, most countries increased network capacity rapidly (OECD, 2021a), suggesting supply was not binding.

- **Measurement timing:** Annual data may miss within-year dynamics. If price and quantity adjusted at different times within years, relationships could appear weak. Higher-frequency (monthly) data would clarify dynamics, though cross-country monthly data are unavailable.

- **Government interventions:** Some countries implemented broadband subsidies or price controls during COVID (OECD, 2021a). If subsidies offset price increases, observed relationships could be distorted. Detailed policy data would help control for interventions.

Despite these limitations, the robustness across specifications, price measures (Table A.3), and time periods provides confidence in core findings.

7. Conclusion

This paper documents a fundamental transformation in broadband demand across 33 European countries over 2010–2024. Three findings emerge from our two-way fixed effects analysis with Driscoll–Kraay standard errors.

First, substantial regional heterogeneity characterised pre-COVID demand: EaP countries exhibited strong price sensitivity ($\varepsilon = -0.60$, p<0.001)—5.9 times larger than EU elasticity ($\varepsilon = -0.10$, p<0.10)—reflecting income-driven affordability constraints. Second, both regions converged to near-zero elasticity during 2020–2024, with price changes having no detectable effect on adoption—evidence that demand-side digital harmonisation between EaP and EU markets has already been achieved. Third, and most critically, the pre-COVID trend is detectable before 2020: for EU countries, year-by-year estimates show elasticity declining from -0.14 in 2015 toward zero; for EaP countries, the pooled placebo test confirms the same direction (p=0.086), as EaP's small sample (five to six countries) precludes reliable annual estimates. Together, these findings indicate the pandemic accelerated but did not initiate broadband's integration into economic and social life. Methodologically, we demonstrate that price measurement critically affects inference—income-relative prices yield robust results while PPP-adjusted prices do not.

For policy, the near-zero elasticity implies that affordability interventions—effective when demand was elastic throughout the pre-COVID period—have lost traction. Policy must now prioritize infrastructure deployment and uni-

versal service obligations. Methodologically, the finding that only income-relative prices yield robust elasticity estimates implies that policymakers tracking broadband affordability should use GNI-based metrics rather than PPP-adjusted or nominal prices. Broadband increasingly resembles utilities like water or electricity, potentially justifying stronger regulatory oversight and public investment to ensure equitable access.

Future research should examine household-level heterogeneity to identify remaining price-sensitive populations, extend analysis to mobile broadband and other regions, and develop quality-adjusted price indices. As broadband completes its transformation into essential infrastructure, understanding these dynamics becomes crucial for bridging digital divides worldwide.

CRediT authorship contribution statement

Samir Orujov: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Ilgar Ismayilov:** Validation, Writing – review & editing, Supervision. **Jeyhun Huseynzade:** Validation, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study are publicly available from the International Telecommunication Union (ITU) World Telecommunication/ICT Indicators Database and the World Bank World Development Indicators. Replication code and processed analysis datasets are available at: https://github.com/sorujov/Broadband-Demand-Elasticity.

Appendix A. Regression Tables

This appendix presents the complete regression results referenced in the main text. All models use two-way fixed effects (country and year) with Driscoll–Kraay standard errors robust to heteroskedasticity, serial correlation, and cross-sectional dependence (bandwidth = 3).

Appendix A.1. Baseline Pre-COVID Results

Table A.1 presents the baseline estimates for the pre-COVID period (2010–2019) across seven control specifications.

Table A.1: Baseline Two-Way Fixed Effects Estimates: Pre-COVID Period (2010–2019)

	Dependent Variable: Log(Subscriptions per 100)						
	(1) GDP Only	(2) + Socio	(3) + Instit.	(4) + Infra.	(5) + Demog.	(6) + All	(7) Full
Panel A: Price Elasticity							
Log(Price)	−0.08*	−0.14***	−0.15***	−0.10*	−0.09	−0.09	−0.10*
	(0.04)	(0.04)	(0.04)	(0.06)	(0.06)	(0.06)	(0.05)
Log(Price) × EaP	−0.53***	−0.48***	−0.46***	−0.48***	−0.48***	−0.48***	−0.49***
	(0.08)	(0.05)	(0.05)	(0.07)	(0.07)	(0.07)	(0.05)
Implied EaP elasticity	−0.62***	−0.62***	−0.61***	−0.58***	−0.57***	−0.57***	−0.60***
	(0.08)	*(0.08)*	*(0.07)*	*(0.07)*	*(0.07)*	*(0.07)*	*(0.06)*
Panel B: Control Variables							
Log(GDP per capita)	0.19***	0.22***	0.21***	0.22***	0.27***	0.27***	0.23***
	(0.07)	(0.07)	(0.06)	(0.08)	(0.08)	(0.08)	(0.07)
Urban population (%)		0.01	0.01	0.01	0.01	0.01	0.01
		(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Tertiary enrollment (%)		0.01***	0.01***	0.01***	0.01***	0.01***	0.01***
		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Regulatory quality			0.06**	0.02	0.04	0.04	0.03
			(0.03)	(0.04)	(0.03)	(0.03)	(0.03)
Log(Secure servers)				0.05**	0.08**	0.08**	0.07**
				(0.02)	(0.03)	(0.03)	(0.03)
R&D (% GDP)				0.03	0.05**	0.05**	0.04**
				(0.03)	(0.02)	(0.02)	(0.02)
Log(Population density)					0.89***	0.89***	0.84***
					(0.19)	(0.19)	(0.17)
Working-age pop. (15–64, %)					0.01	0.01	0.01
					(0.01)	(0.01)	(0.01)
Panel C: Model Statistics							
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	330	330	330	330	330	330	330
Countries	33	33	33	33	33	33	33
R-squared	0.27	0.34	0.34	0.37	0.39	0.39	0.40

Notes: Dependent variable is log fixed broadband subscriptions per 100 inhabitants. Price measured as % of GNI per capita. EaP dummy for Eastern Partnership countries. Columns (6)–(7) include all non-macro controls; (7) adds GDP growth and inflation (coefficients not shown). Country and year FE. Driscoll–Kraay SEs (bandwidth = 3). Qualitatively unchanged with clustered or robust SEs. *p<0.10, **p<0.05, ***p<0.01.

Appendix A.2. COVID-19 Interaction Effects

Table A.2 presents results from the full sample (2010–2024) with COVID-19 period interactions.

Table A.2: COVID-19 Interaction Effects: Full Sample (2010–2024)

	Dependent Variable: Log(Subs. per 100)			
	(1) Full Sample w/ COVID Int.	(2) Pre-COVID 2010–19	(3) COVID 2020–24	(4) Diff. Test (p-value)
Panel A: EU Countries				
Log(Price)	−0.19**	−0.10*	−0.04**	–
	(0.07)	(0.05)	(0.02)	
Log(Price) × COVID	+0.30***	–	–	0.007
	(0.11)			
Implied COVID elasticity	+0.11	–	−0.04**	–
	(0.07)		(0.02)	
Panel B: EaP Countries				
Log(Price) × EaP	−0.17*	−0.60***	–	–
	(0.10)			
Log(Price) × EaP × COVID	+0.12***	–	–	0.000
	(0.03)			
Implied pre-COVID EaP elasticity	−0.36***	−0.60***	–	–
	(0.11)	(0.06)		
Implied COVID EaP elasticity	+0.06	–	−0.13	–
	(0.12)		(0.10)	
Panel C: Change in Elasticity				
$\Delta\varepsilon_{EU}$ (COVID – Pre)	+0.30***	–	–	0.007
	(0.11)			
$\Delta\varepsilon_{EaP}$ (COVID – Pre)	+0.42***	–	–	0.002
	(0.14)			
Panel D: Model Statistics				
Full controls	Yes	Yes	Yes	–
Country FE	Yes	Yes	Yes	–
Year FE	Yes	Yes	Yes	–
Observations	495	330	165	–
Countries	33	33	33	–
R-squared	0.42	0.40	0.40	–

Notes: Dependent variable is log of fixed broadband subscriptions per 100 inhabitants. Price is measured as percentage of GNI per capita. COVID is a period dummy for years 2020–2024. EaP is a dummy for Eastern Partnership countries. Full controls include: log GDP per capita, urban population %, tertiary enrollment %, regulatory quality, log secure servers, R&D % GDP, log population density, and working-age population share (15–64, %). Column (1) presents the full interaction model on 2010–2024 data. Columns (2)–(3) present separate subsample estimates for comparison. Column (4) reports p-values from Wald tests of coefficient differences between COVID and pre-COVID periods. Driscoll–Kraay standard errors (bandwidth = 3) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix A.3. Robustness to Alternative Price Measures

Table A.3 compares elasticity estimates across three price definitions: income-relative (baseline), PPP-adjusted, and nominal USD.

Table A.3: Price Elasticity Robustness: Alternative Price Measures and Control Specifications (Pre-COVID)

Price Measure	Control Spec	EU Elast.	EU SE	EU p	EaP Elast.	EaP SE	EaP p	N	R^2
Price: GNI%									
	Full Controls (Baseline)	−0.101*	(0.053)	0.056	−0.595***	(0.061)	0.000	330	0.40
	Comprehensive	−0.101*	(0.054)	0.063	−0.570***	(0.074)	0.000	330	0.36
	Core	−0.142***	(0.042)	0.001	−0.618***	(0.075)	0.000	330	0.34
	Institutional	−0.090**	(0.043)	0.040	−0.604***	(0.073)	0.000	330	0.28
	Infrastructure	−0.033	(0.074)	0.655	−0.596***	(0.075)	0.000	330	0.30
	Demographic	−0.104***	(0.028)	0.000	−0.621***	(0.072)	0.000	330	0.28
	Macroeconomic	−0.094**	(0.040)	0.019	−0.631***	(0.057)	0.000	330	0.30
	Minimal	−0.082*	(0.042)	0.050	−0.615***	(0.075)	0.000	330	0.27
Price: PPP									
	Full Controls (Baseline)	−0.087*	(0.052)	0.094	−0.201*	(0.116)	0.084	330	0.22
	Comprehensive	−0.092*	(0.055)	0.095	−0.211*	(0.108)	0.051	330	0.19
	Core	−0.125**	(0.063)	0.048	−0.244**	(0.106)	0.021	330	0.13
	Institutional	−0.116**	(0.052)	0.027	−0.164	(0.113)	0.149	330	0.07
	Infrastructure	−0.081	(0.050)	0.107	−0.155	(0.109)	0.158	330	0.09
	Demographic	−0.123*	(0.063)	0.052	−0.180	(0.112)	0.111	330	0.06
	Macroeconomic	−0.104**	(0.048)	0.031	−0.168	(0.124)	0.178	330	0.08
	Minimal	−0.112**	(0.051)	0.030	−0.177	(0.111)	0.112	330	0.05
Price: USD									
	Full Controls (Baseline)	−0.098**	(0.039)	0.014	−0.340***	(0.059)	0.000	330	0.29
	Comprehensive	−0.099**	(0.039)	0.012	−0.337***	(0.063)	0.000	330	0.26
	Core	−0.124***	(0.042)	0.003	−0.390***	(0.063)	0.000	330	0.23
	Institutional	−0.100***	(0.031)	0.001	−0.332***	(0.065)	0.000	330	0.14
	Infrastructure	−0.064	(0.041)	0.120	−0.326***	(0.059)	0.000	330	0.16
	Demographic	−0.115***	(0.039)	0.004	−0.356***	(0.063)	0.000	330	0.14
	Macroeconomic	−0.090***	(0.030)	0.003	−0.344***	(0.061)	0.000	330	0.16
	Minimal	−0.094***	(0.029)	0.002	−0.348***	(0.066)	0.000	330	0.14

Notes: Each row is a separate regression. All specifications include country and year fixed effects. Driscoll–Kraay standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix A.4. Placebo Test for Pre-COVID Trends

Table A.4 presents the placebo test splitting the pre-COVID period into early (2010–2014) and late (2015–2019) subperiods.

Table A.4: Placebo Test: Pre-COVID Trends (2010–2019)

	Dependent Variable: Log(Subs. per 100)		
	(1) Full Sample 2010–19	(2) Early 2010–14	(3) Late 2015–19
Panel A: EU Countries			
Log(Price)	-0.15^{*}	$+0.21$	-0.06^{***}
	(0.09)	(0.14)	(0.02)
Log(Price) × Late	$+0.13^{*}$	–	–
	(0.07)		
Implied late elasticity	-0.02	–	-0.06^{***}
	(0.06)		(0.02)
Panel B: EaP Countries			
Log(Price) × EaP	-0.29^{***}	-0.33^{***}	-0.08
	(0.10)	(0.02)	(0.08)
Log(Price) × EaP × Late	$+0.08^{*}$	–	–
	(0.04)		
Implied early EaP elasticity	-0.44^{***}	-0.33^{***}	–
	(0.03)	(0.02)	
Implied late EaP elasticity	-0.23^{**}	–	-0.08
	(0.11)		(0.08)
Panel C: Change in Elasticity			
$\Delta\varepsilon_{EU}$ (Late – Early)	$+0.13^{*}$	–	–
	(0.07)		
p-value	0.057	–	–
$\Delta\varepsilon_{EaP}$ (Late – Early)	$+0.08^{*}$	–	–
	(0.04)		
p-value	0.086	–	–
Panel D: Model Statistics			
Full controls	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	330	165	165
Countries	33	33	33
R-squared	0.45	0.49	0.21

Notes: Dependent variable: log fixed broadband subscriptions per 100. Price: % of GNI per capita. "Late" is a placebo indicator for 2015–2019. EaP denotes Eastern Partnership countries. Column (1) estimates the full placebo interaction on 2010–2019; Columns (2)–(3) report subsamples for 2010–2014 (early) and 2015–2019 (late). Controls: log GDP per capita, urbanization (%), tertiary enrollment (%), regulatory quality, log secure servers, R&D (% GDP), log population density, and working-age population share (15–64, %). Driscoll–Kraay SEs (bandwidth = 3) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B. Additional Robustness Checks

Appendix B.1. Period-Split: Acute vs. Post-Acute COVID

Table B.1 decomposes the COVID period (2020–2024) into an acute phase (2020–2021) and a post-acute phase (2022–2024). Both sub-periods exhibit near-zero elasticity for EU and EaP countries, confirming that the disappearance of price sensitivity is not driven solely by the acute pandemic shock. Panel B excludes Ukraine and Belarus—whose data reliability is compromised by war and sanctions—and reaches the same conclusion.

Table B.1: Period-Split Elasticity Estimates: Pre-COVID, Acute COVID, and Post-Acute

	Pre-COVID 2010–2019		Acute COVID 2020–2021		Post-Acute 2022–2024	
	EU	EaP	EU	EaP	EU	EaP
Panel A: Full Sample (33 countries)						
Elasticity	−0.197**	−0.359***	0.073	0.020	0.124	0.098
	(0.077)	(0.112)	(0.067)	(0.113)	(0.080)	(0.109)
Observations			495			
R-squared			0.42			
Panel B: Excluding Ukraine & Belarus from COVID Periods						
Elasticity	−0.191***	−0.427***	0.071	−0.080	0.079	−0.051
	(0.067)	(0.096)	(0.058)	(0.099)	(0.068)	(0.103)
Observations			485			
R-squared			0.43			
Country FE			Yes			
Year FE			Yes			
Price measure			GNI% (income-relative)			

Notes: Estimates come from a single regression on the full 2010–2024 panel using price×period interaction dummies for Acute COVID (2020–2021) and Post-Acute (2022–2024). Pre-COVID elasticities are the base-period coefficients. This interaction approach avoids the short-panel problem of estimating 2-year subsamples. Full controls. Driscoll–Kraay standard errors (bandwidth = 3) in parentheses. Panel B excludes Ukraine and Belarus from the COVID periods only (their pre-2020 observations are retained). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B.2. IV Robustness: 2SLS Estimates

Table B.2 presents 2SLS estimates instrumenting fixed broadband price with its own one-year lag (log fixed broadband price in GNI%, lagged one year).

Instrument rationale. National fixed broadband price baskets are set through annual regulatory and procurement cycles, generating high year-to-year autocorrelation. The lagged price therefore strongly predicts the current price (within-demeaned first-stage $F = 65$ pre-COVID; $F = 99$ full-sample), well above the Stock–Yogo 10% maximal-IV-bias threshold of 16.38 (Stock et al., 2005). The exclusion restriction is plausible: conditional on country and year fixed effects plus time-varying controls, past prices affect current subscriptions only through current prices, since broadband contracts are monthly or annual and consumers cannot anticipate next year's regulated tariff when making adoption decisions today.[1]

Table B.2: IV Robustness: 2SLS Estimates Using Lagged Own Price as Instrument

	EU Elasticity		EaP Elasticity		
Specification	Coef.	SE	Coef.	SE	First-Stage F
A. Pre-COVID (2010–2019)					
OLS (TWFE + DK SE)	−0.101*	(0.053)	−0.595***	(0.061)	—
2SLS, lagged price	−0.172*	(0.097)	−0.615***	(0.091)	65
B. Full sample (2010–2024), COVID interactions					
2SLS, lagged price	−0.193***	(0.058)	−0.340***	(0.088)	99

Notes: Instrument: log fixed broadband price (GNI%) lagged one year. All specifications include country and year fixed effects absorbed via two-way within-transformation. First-stage F is a partial F-statistic for the excluded instrument in the within-transformed first stage; $F > 10$ indicates instrument relevance (Stock et al., 2005). EaP elasticity = EU coefficient + EU×EaP interaction; SE via delta method. OLS: Driscoll–Kraay SE (kernel bandwidth=3). 2SLS: heteroskedasticity-robust SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Results. Panel A of Table B.2 shows pre-COVID results alongside the OLS baseline. The 2SLS estimate is EU= −0.172 (SE= 0.097, $p = 0.075$*) and EaP= −0.615 (SE= 0.091, p =0.000***)—broadly consistent with OLS (EU= −0.101,

[1]We also tested international internet bandwidth as an alternative instrument. However, an over-identified specification combining both instruments strongly rejected the Hansen J overidentification test ($\chi^2(2) = 262.89$, $p < 0.001$), indicating that bandwidth correlates with connection speed and quality, which directly enter the demand equation. A further practical limitation is that ITU bandwidth data are missing for most EU countries after 2018, and multiple imputation does not recover sufficient coverage to support credible inference. Lagged price is therefore the sole preferred instrument.

EaP= -0.595), confirming that upward simultaneity bias in the baseline TWFE estimates is negligible. The EaP 2SLS and OLS estimates are nearly identical; the EU 2SLS coefficient is somewhat larger in magnitude but within sampling variance, consistent with minor attenuation bias from price measurement error rather than simultaneity. Panel B uses the same instrument in the full-sample specification with COVID interactions: EU= -0.193 (SE= 0.058, p =0.001***) and EaP= -0.340 (SE= 0.088, p =0.000***), consistent with the COVID-era attenuation of price sensitivity found in OLS.

Overall, the IV results directly validate the OLS estimates: the endogeneity concern is resolved and no evidence of upward simultaneity bias is found.

Appendix B.3. EaP Jackknife: Leave-One-Country-Out

Table B.3 reports leave-one-out estimates dropping each of the six EaP countries in turn. EaP elasticity ranges from -0.42 to -0.67 across the six sub-samples. Armenia is the most influential country: its exclusion raises the EaP estimate from -0.60 to -0.42, suggesting Armenia contributes relatively high price-sensitivity to the aggregate EaP estimate. Excluding Georgia yields the largest negative estimate (-0.67), indicating that Georgia exhibits below-average price sensitivity relative to the rest of the EaP group; removing it therefore shifts the group-level estimate further from zero. Crucially, *all* leave-one-out EaP estimates are statistically significant at the 1% level, and all confirm a substantial, negative price elasticity. The EaP result is therefore not driven by any single country, and the range $[-0.42, -0.67]$ reflects genuine heterogeneity within the EaP group rather than measurement fragility.

Table B.3: EaP Jackknife Robustness: Leave-One-Country-Out (Pre-COVID Baseline)

	EU Elasticity		EaP Elasticity			
Sample	Coef.	SE	Coef.	SE	N	R^2
Full EaP (baseline)	-0.101^{*}	(0.053)	-0.595^{***}	(0.061)	330	0.40
Drop Armenia	-0.069	(0.059)	-0.417^{***}	(0.040)	320	0.34
Drop Azerbaijan	-0.092^{*}	(0.048)	-0.548^{***}	(0.058)	320	0.41
Drop Belarus	-0.089	(0.062)	-0.667^{***}	(0.074)	320	0.48
Drop Georgia	-0.087^{**}	(0.037)	-0.672^{***}	(0.068)	320	0.49
Drop Moldova	-0.100^{*}	(0.053)	-0.604^{***}	(0.062)	320	0.39
Drop Ukraine	-0.107^{**}	(0.049)	-0.604^{***}	(0.069)	320	0.41

Notes: Each row drops one EaP country from the sample and re-estimates the baseline model (Full Controls, GNI% price, pre-COVID 2010–2019). Driscoll–Kraay standard errors (bandwidth = 3) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B.4. Sample Restrictions

Table B.4 reports robustness to alternative sample definitions. The balanced-panel restriction (countries with complete 2010–2019 data) yields identical estimates to the full sample, confirming that attrition is not driving the results. Excluding outliers identified by Cook's distance ($d > 4/N$) removes 22 observations across 2 countries with negligible impact on either elasticity estimate. For the high-income subsample (17 EU countries above median GDP per capita), no EaP countries are present, so only the EU elasticity (−0.016, p=0.673) is identified; this estimate is not statistically significant, consistent with higher-income consumers facing minimal affordability constraints. For the middle/lower-income subsample (16 countries including all EaP), the EaP elasticity (−0.433, p<0.001) is somewhat smaller than the full-sample estimate, reflecting country heterogeneity at lower income levels.

Table B.4: Sample Restriction Robustness Checks (Pre-COVID Baseline, Full Controls, GNI%-price). Driscoll–Kraay standard errors in parentheses. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	EU		EaP			
Sample Restriction	$\hat{\varepsilon}$	SE	$\hat{\varepsilon}$	SE	N	Countries
Full sample (baseline)	−0.101*	(0.053)	−0.595***	(0.061)	330	33
Balanced panel only	−0.101*	(0.053)	−0.595***	(0.061)	330	33
Excl. outliers (Cookś $d > 4/N$)	−0.104*	(0.057)	−0.594***	(0.062)	308	31
High-income countries only	−0.016	(0.037)	—		170	17
Middle/lower-income only	−0.008	(0.120)	−0.433***	(0.089)	160	16

Notes: All models estimated with two-way fixed effects (country and year) and Driscoll–Kraay standard errors (bandwidth = 3). Price variable is log fixed broadband price as % of GNI per capita. High/middle-income split uses median country-level GDP per capita (constant 2015 USD) averaged over 2010–2019. Cookś distance threshold is $4/N$ where N is the number of observations.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) for programming and data-collection scripting, and for manuscript editing. After using this tool, the authors reviewed and edited all content as needed and take full responsibility for the content of this publication.

References

Baltagi, B.H., 2021. Econometric analysis of panel data. 6th ed., Springer, Cham, Switzerland.

Becker, G.S., Murphy, K.M., 1988. A theory of rational addiction. Journal of Political Economy 96, 675–700.

Bertschek, I., Niebel, T., 2016. The drivers of digital transformation of business: The role of human capital, infrastructure, and firm age. Information Economics and Policy 37, 52–64.

Bökermann, M., Scheuermann, B., 2021. COVID-19 pandemic and internet traffic: A longitudinal study. Computer Networks 192, 108098.

Brynjolfsson, E., Horton, J.J., Ozimek, A., Rock, D., Sharma, G., TuYe, H.Y., 2020. COVID-19 and remote work: An early look at US data. NBER Working Paper 27344. National Bureau of Economic Research.

Brynjolfsson, E., Hu, Y.J., Smith, M.D., 2003. Consumer surplus in the digital economy: Estimating the value of increased product variety at online booksellers. Management Science 49, 1580–1596.

Callaway, B., Sant'Anna, P.H.C., 2021. Difference-in-differences with multiple time periods. Journal of Econometrics 225, 200–230.

Cameron, A.C., Miller, D.L., 2015. A practitioner's guide to cluster-robust inference. Journal of Human Resources 50, 317–372.

de Chaisemartin, C., D'Haultfœuille, X., 2020. Two-way fixed effects estimators with heterogeneous treatment effects. American Economic Review 110, 2964–2996.

Czernich, N., Falck, O., Kretschmer, T., Woessmann, L., 2011. Broadband infrastructure and economic growth. Economic Journal 121, 505–532.

Deaton, A., Muellbauer, J., 1980. Economics and consumer behavior. Cambridge University Press, Cambridge, UK.

Dietrich, A.M., Keuster, K., Muller, G.J., Schoenle, R., 2020. Consumption patterns during the COVID-19 pandemic. ECB Working Paper Series 2448. European Central Bank.

Driscoll, J.C., Kraay, A.C., 1998. Consistent covariance matrix estimation with spatially dependent panel data. Review of Economics and Statistics 80, 549–560.

Ecorys, 2013. Evaluation of the EU broadband policy. Final Report. European Commission. Brussels.

European Commission, 2021. Eastern Partnership: Connecting to digital opportunities. Policy Brief. European Commission. Brussels.

Favale, T., Soro, F., Trevisan, M., Drago, I., Mellia, M., 2020. Campus traffic and e-learning during COVID-19 pandemic. Computer Networks 176, 107290.

Goldfarb, A., Tucker, C., 2019. Digital economics. Journal of Economic Literature 57, 3–43.

Goodman-Bacon, A., 2021. Difference-in-differences with variation in treatment timing. Journal of Econometrics 225, 254–277.

Goolsbee, A., Klenow, P.J., 2006. Valuing consumer products by the time spent using them: An application to the internet. American Economic Review Papers and Proceedings 96, 108–113.

Greenstein, S., Mazzeo, M.J., 2016. Measuring the quality of communication networks, in: Goldfarb, A., Greenstein, S.M., Tucker, C.E. (Eds.), Economic analysis of the digital economy. University of Chicago Press, Chicago, IL, pp. 321–346.

Greenstein, S., McDevitt, R.C., 2016. The digital economy and the modern economy: A review of the literature. Review of Industrial Organization 49, 289–321.

Gruber, H., Koutroumpis, P., 2014. The diffusion of mobile telecommunications services in the EU. European Economic Review 70, 262–282.

Grzybowski, L., 2015a. Estimating broadband diffusion in the EU using NUTS 1 regional data. Telecommunications Policy 39, 798–810.

Grzybowski, L., 2015b. Fixed broadband services diffusion in the EU-27 member states. Telecommunications Policy 39, 810–821.

Grzybowski, L., Nitsche, R., Verboven, F., Wiethaus, L., 2014. Market definition for broadband internet in Slovakia—are fixed and mobile technologies in the same market? Information Economics and Policy 28, 39–56.

Guermazi, B., Satola, D., 2021. Broadband for all: A holistic approach. Working Paper. World Bank. Washington, DC.

Guermazi, B., Satola, D., Rastogi, S., 2021. Digital infrastructure for development: From principles to practice. Digital Development Report. World Bank. Washington, DC.

Hauge, J.A., Prieger, J.E., 2010. Demand-side programs for broadband. Telecommunications Policy 34, 470–479.

Hausman, J.A., Sidak, J.G., Singer, H.J., 2001. Private incentives and social costs: Estimating the demand for broadband services. Yale Journal on Regulation 18, 227–304.

Hoechle, D., 2007. Robust standard errors for panel regressions with cross-sectional dependence. Stata Journal 7, 281–312.

IAB, 2020. Internet trends during COVID-19 pandemic: A global perspective. Technical Report. Internet Architecture Board.

ITU, 2024. World telecommunication/ICT indicators database. Technical Report. International Telecommunication Union. Geneva. Available at: https://www.itu.int/en/ITU-D/Statistics/.

Katz, M.L., Shapiro, C., 1985. Network externalities, competition, and compatibility. American Economic Review 75, 424–440.

Katz, R.L., Koutroumpis, P., Callorda, F.M., 2010. The impact of broadband on the economy: Research to date and policy issues. ITU Study 4, 1–35.

Kolko, J., 2012. Broadband and local growth. Journal of Urban Economics 71, 100–113.

Koutroumpis, P., 2009. The economic impact of broadband on growth: A simultaneous approach. Telecommunications Policy 33, 471–485.

Laffont, J.J., Tirole, J., 2000. Competition in telecommunications. MIT Press, Cambridge, MA.

Lindlacher, V., 2021. Low demand despite broad supply: Is high-speed Internet an infrastructure of general interest? Information Economics and Policy 56, 100924.

Liu, Y.H., Prince, J., Wallsten, S., 2018. Distinguishing bandwidth and latency in households' willingness-to-pay for broadband internet speed. Information Economics and Policy 45, 1–15.

Madden, G., Coble-Neal, G., 2015. Demand estimation and optimal pricing for communications services. Telecommunications Policy 39, 231–243.

McKinsey & Company, 2020. How COVID-19 has pushed companies over the technology tipping point. Survey Report. McKinsey & Company.

OECD, 2020. Broadband connectivity: An agenda for policy action. OECD Digital Economy Papers 295. Organisation for Economic Co-operation and Development. Paris.

OECD, 2021a. Broadband policy responses to the COVID-19 pandemic. OECD Digital Economy Papers 317. Organisation for Economic Co-operation and Development. Paris.

OECD, 2021b. The impact of COVID-19 on broadband: Getting ready for the next wave. OECD Digital Economy Papers 320. Organisation for Economic Co-operation and Development. Paris.

Petersen, M.A., 2009. Estimating standard errors in finance panel data sets: Comparing approaches. Review of Financial Studies 22, 435–480.

Rappoport, P.N., Kridel, D.J., Taylor, L.D., Alleman, J.H., Duffy-Deno, K.T., 2003. Demand for access to the internet. Internet economics , 55–72.

Röller, L.H., Waverman, L., 2001. Telecommunications infrastructure and economic development: A simultaneous approach. American Economic Review 91, 909–923.

Rubin, D.B., 1987. Multiple imputation for nonresponse in surveys. John Wiley & Sons, New York.

Schreyer, P., 2002. OECD manual: Measuring the information economy. Technical Report. OECD Publishing. Paris.

Sinclair, A., 2023. Assessing the benefits of high-speed broadband: Lessons from Australia's National Broadband Network (NBN). Information Economics and Policy 65, 101062.

Stock, J.H., Wright, J.H., Yogo, M., 2005. Testing for weak instruments in linear IV regression, in: Andrews, D.W.K., Stock, J.H. (Eds.), Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg. Cambridge University Press, Cambridge, pp. 80–108.

Wallsten, S.J., 2006. Broadband and unbundling regulations in OECD countries. Journal of Regulatory Economics 30, 313–326.

Waverman, L., Meschi, M., Fuss, M., 2001. Telecommunications infrastructure and economic development: A simultaneous approach. American Economic Review Papers and Proceedings 91, 384–388.

Wooldridge, J.M., 2010. Econometric analysis of cross section and panel data. 2nd ed., MIT Press, Cambridge, MA.

World Bank, 2020. International comparison program (ICP). Available at: https://www.worldbank.org/en/programs/icp.

World Bank, 2024a. World development indicators. Available at: https://databank.worldbank.org/source/world-development-indicators.

World Bank, 2024b. Worldwide governance indicators. Available at: https://www.worldbank.org/en/publication/worldwide-governance-indicators.